DATA PREPROCESSING

What is data ?

Data is in format TEXT,Image,Audio,Number

Data Preprocessing is a process to convert raw data into meaningfull data using different technique.

Why data preprocessing is Important

Data in the real world is dirty.

Steps in data preprocessing

1 data cleaning

2. data integration

3. data reduction

4. data transformation

5. data discretization

What is Data Cleaning?

- **Data Cleaning means fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data.**

CO_2 emissions (metric tons per capita)

Country Name	2000	2001	2002
United States	20.17875	19.63651	NAN
United Kingdom	9.199549	9.233175	8.904123
India	0.97987	NAN	0.967381
China	2.696862	2.742121	3.007083
Russian Federation	NAN	10.6696	10.7159
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What is Data Integration?

- **Data Integration** is a technique to merges data from multiple sources into a coherent data store, such as a data warehouse.

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What is Data Reduction?

- **Data Reduction** is a technique use to reduce the data size by aggregating, eliminating redundant features, or clustering, for instance.

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What is Data Transformation?

- **Data Transformation** means data are transformed or consolidated into forms appropriate for ML model training, such as normalization, may be applied where data are scaled to fall within a smaller range like 0.0 to 1.0.
  - Aggregation
  - Feature type conversion:
  - Normalization
  - Attribute/feature construction

Country Code	2000	2002
USA	0.20	0.19
GBR	0.09	0.08
IND	0.01	0.09
CHN	0.02	0.03
RUS	0.10	0.10
AUS	0.17	0.17

What is Data Discretization?

- **Data Discretization** technique transforms numeric data by mapping values to interval or concept labels
- It can be used to reduce the number of values for a given continuou attribute by dividing the range of the attribute into intervals.
- Discretization techniques include -
  - Binning
  - Histogram analysis
  - Cluster analysis
  - Decision-tree analysis

Ex-
Age: 1,2,3,4,5,6,7,8,9
Output>>>
Age: 1-3, 4-6, 7-9

What is Feature Engineering?

- **Feature** is an attribute or property shared by all of the independent units on which analysis or prediction is to be done.
- **Feature Engineering** is the process to create feature/extract the feature from existing features by domain knowledge to increase the performance of Machine Learning Model.
- Feature Engineering is an art.

Why is Feature Engineering Important?

- Quality data always help to improve the accuracy and performance of Machine Learning model.
- Machine Learning algorithm follow the rule *(learn like Kids)*
  - GIGO – Garbage In Garbage Out

Examples of Feature Engineering?

Train Boarding Time	Train Reach at Station	Delayed or On Time	Delay Time in Minute
10.00 AM	10.15 AM	Delayed	00.15
05.00 PM	05.00 PM	On time	00.00

Examples of Feature Engineering?

DateTime	Hour	Day	Month	Year	Day of Week
07-Apr-2020 12.00.00	12	7	4	2020	2
10-Apr-2020 23.04.00	23	8	4	2020	5

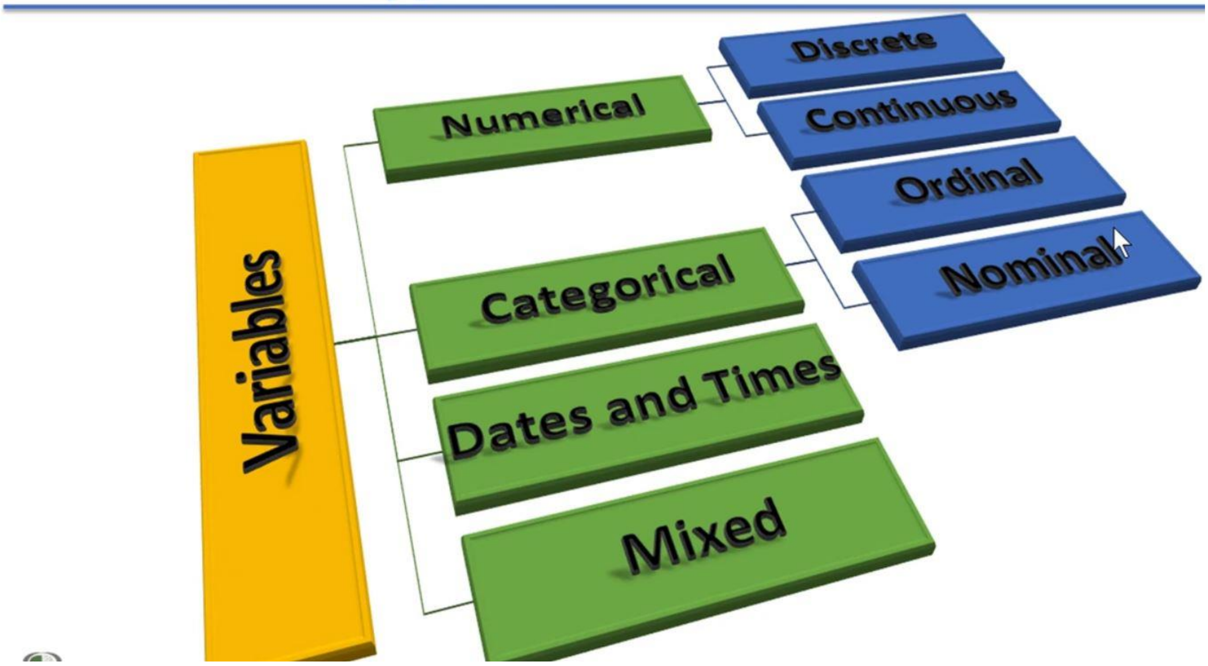
Examples of Feature Engineering?

Age	Sex	Male	Female	Age range
23	Male	1	0	21-30
12	Female	0	1	11-20
34	Male	1	0	31-40
26	Female	0	1	21-30
15	Male	1	0	11-20
38	Male	1	0	31-40

What is Variable?

- A **variable** is any characteristic, number, or quantity that can be measured or counted.

What are the type of variable?



What is Numerical Variable ?

- **A numerical variable is a variable where the measurement or number has a numerical meaning.**

Age	Weight in KG	Income	No. of Vehicles	Percentage
23	92.2	59,00,000	2	90.83
12	41.3	5000	1	62.00
34	80.9	3,50,000	5	65.00
26	78.0	10,00,000	1	80.01

What is Discrete Variable?

- The value which can be counted as whole value(fixed) called as **discrete variable**.
- Ex. 1, 45, 73, 23 not 23.4, 56.76, 3.04
- Integer Value

Age	Income	No. of Vehicles
23	59,00,000	2
12	5,000	1
34	3,50,000	5
26	10,00,000	1
15	10,000	1
38	89,16,000	3

What is Continuous Variable?

- The value contain range, measurements called as **continuous variable**.
- Ex. 1.3, 4.5, 73.67, 20.3 not 23, 56, 3
- Floating Value

Weight in KG	Temperature	Percentage
92.2	45.30	90.83
41.3	101.00	62.00
80.9	43.00	65.00
78.0	36.9	80.01
35.9	10.5	72.06
105.0	71.82	99.99

What is Categorical Variable ?

- **The values of a categorical variable are selected from a group of categories, also called labels.**

Grade	Quality	Days	Products	Language
A	Poor	Monday	Laptop	Python
B	Good	Tuesday	Bag	Java
C	Very Good	Wednesday	Mouse	C++
D	Excellent	Thursday	Torch	C
E	Extreme Excellent	Friday	Mobile	Go
F	EE ka Bap	Saturday	Shirt	HTML

What is Ordinal Variables?

- **Ordinal variables are categorical variable in which the categories can be meaningfully ordered.**
- **When the ordinal variable convert into the number then it has mathematical value.**

Grade	Quality	Moths
A	Poor	Jan
B	Good	Feb
C	Very Good	Mar
D	Excellent	Apr
E	Extreme Excellent	May

What is Nominal Variables?

- **Nominal variable** is same like ordinal variable but there is nothing that indicates an intrinsic order of the labels and in principle.
- All labels are equal no order.
- No mathematical value
- Ex. Colour: Red, Green, White, Black

Products	Language
Laptop	Python
Bag	Java
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Mobile	Go
Shirt	HTML

What are the Date and Time Variables?

- **DateTime variables** can contain date only, time only, or date and time.

What are Mixed Variables?

- The variables which contains numbers and categories(labels) called as **Mixed Variables**.